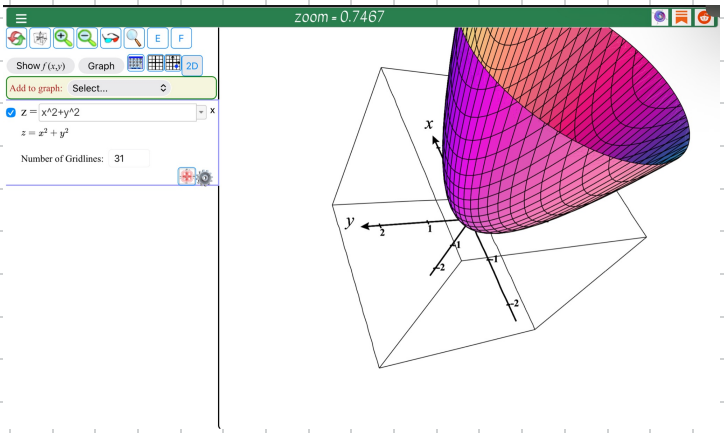
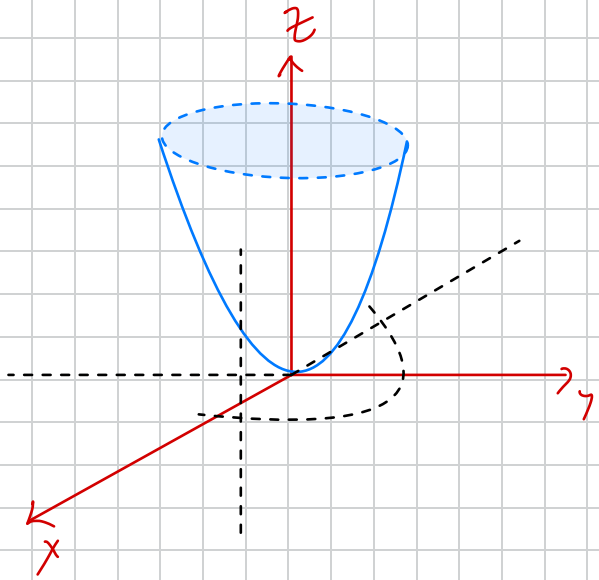
Ejemplo: Resolver la integral que representa el volumen de un sólido. Describir el sólido (gráfica).

$$V = \int_{-2}^{1} \int_{x-1}^{1-x^2} (x^2+y^2)\, dy\, dx$$

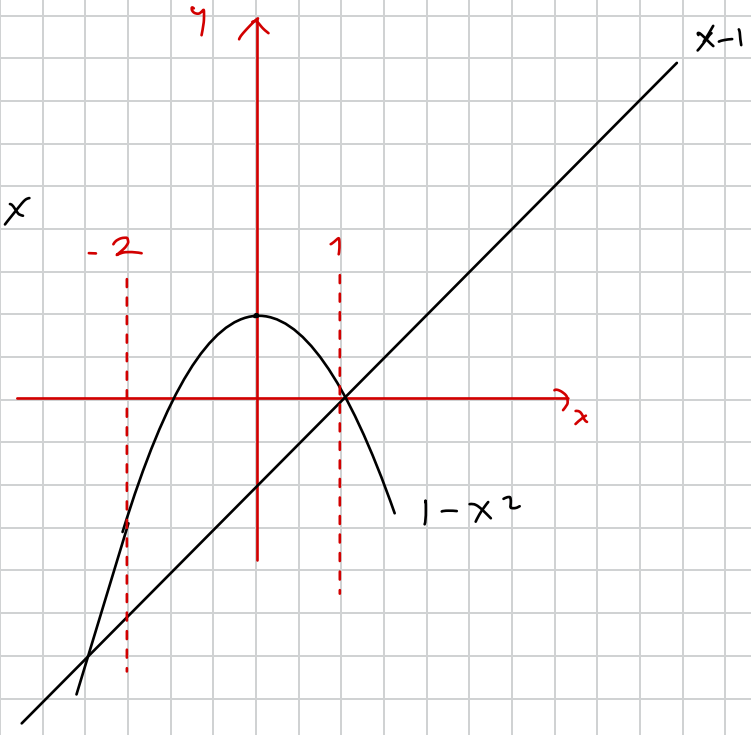
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$$V = \iint_{(I)} f(x,y)\, dy\, dx = \iint_{(II)} f(x,y)\, dx\, dy$$





$$\int_{-2}^{1} \int_{x-1}^{1-x^2} (x^2+y^2)\, dy\, dx$$



$y = x-1$

$y = 1-x^2$

$y-1 = -x^2$

$$V = \int_{-2}^{1} \int_{x-1}^{1-x^2} (x^2+y^2)\, dy\, dx$$

$$V = \int_{-2}^{1} \left(x^2 y + \frac{y^3}{3} \right) \Big|_{x-1}^{1-x^2} dx$$

$$V = \int_{-2}^{1} x^2 \underbrace{(1 - x^2 - x + 1)}_{(2 - x - x^2)} + \frac{1}{3}\left((1-x^2)^3 - (x-1)^3\right) dx$$

$$V = \int_{-2}^{1} \left(2x^2 - x^3 - x^4 + \frac{1}{3}(1-x^2)^3 - \frac{1}{3}(x-1)^3\right) dx$$

$u = x - 1 \quad \frac{1}{3}\int u^3 du$

$du = dx$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$(1-x^2)^3 = 1 - 3x^2 + 3x^4 - x^6$$

$$V = \left(\frac{2}{3}x^3 - \frac{x^4}{4} - \frac{x^5}{5} - \frac{1}{12}(x-1)^4 + \frac{1}{3}\left(x - x^3 + \frac{3}{5}x^5 - \frac{x^7}{7}\right)\right) \Big|_{-2}^{1}$$

$$\frac{2}{3} - \frac{1}{4} - \frac{1}{5} + \frac{1}{3}\left(1 - 1 + \frac{3}{5} - \frac{1}{7}\right)$$

$$\frac{31}{84}$$

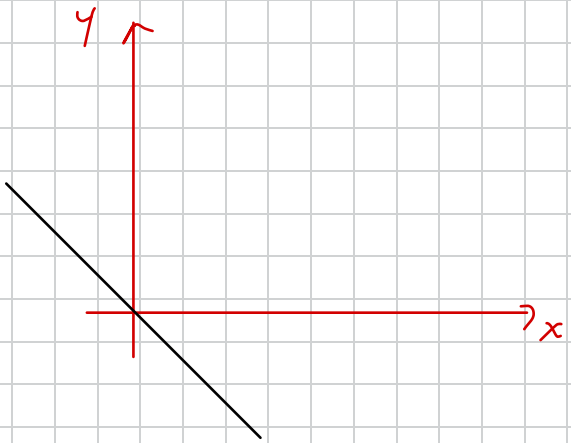
$$-\frac{16}{3} - 4 + \frac{32}{5} - \frac{81}{12} + \frac{1}{3}\left(-2 + 8 - \frac{(3)(32)}{5} + \frac{128}{7}\right)$$

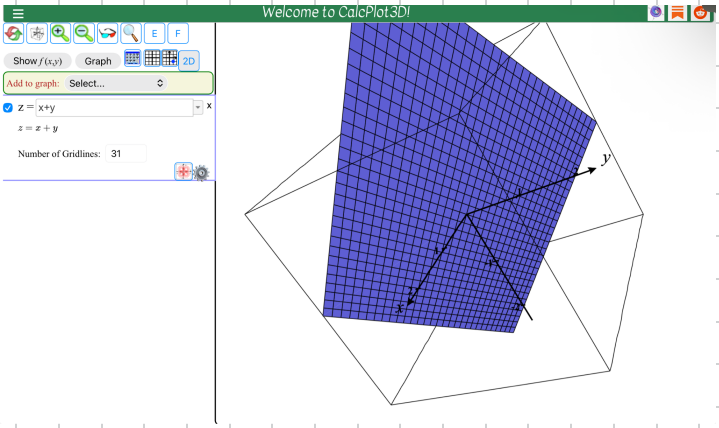
$$-\frac{671}{84} =$$

$$\frac{31}{84} - \left(-\frac{671}{84}\right) = \frac{702}{84} = \frac{117}{14}\ ud^3$$

Ejemplo: Resolver la integral que representa el volumen de un sólido. Describir el sólido (gráfica).

$$V = \int_0^4 \int_{y/4}^{\sqrt{y}} (x+y)\,dx\,dy$$

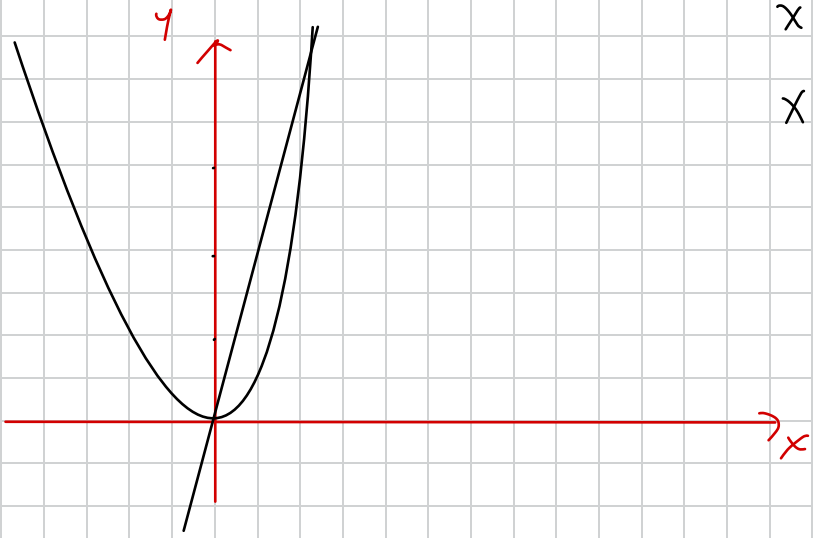




$x = \sqrt{y}$

$x^2 = y$

$x = \frac{y}{4}$; $y = 4x$



$$V = \int_0^4 \int_{y/4}^{\sqrt{y}} (x+y)\,dx\,dy = \int_0^4 \left(\frac{x^2}{2} + xy \right)\Big|_{y/4}^{\sqrt{y}} dy$$

$$= \int_0^4 \left(\frac{y}{2} + y^{3/2} - \frac{9y^2}{32} \right) dy = \frac{y^2}{4} + \frac{2y^{5/2}}{5} - \frac{3}{32} y^3 \Big|_0^4$$

$$= \frac{(4)^2}{4} + \frac{2}{5}(4)^{5/2} - \frac{3}{32}(4)^3$$

$$= \frac{54}{5}\ ud^3$$

Cambio de marco de referencia: Coordenadas polares.

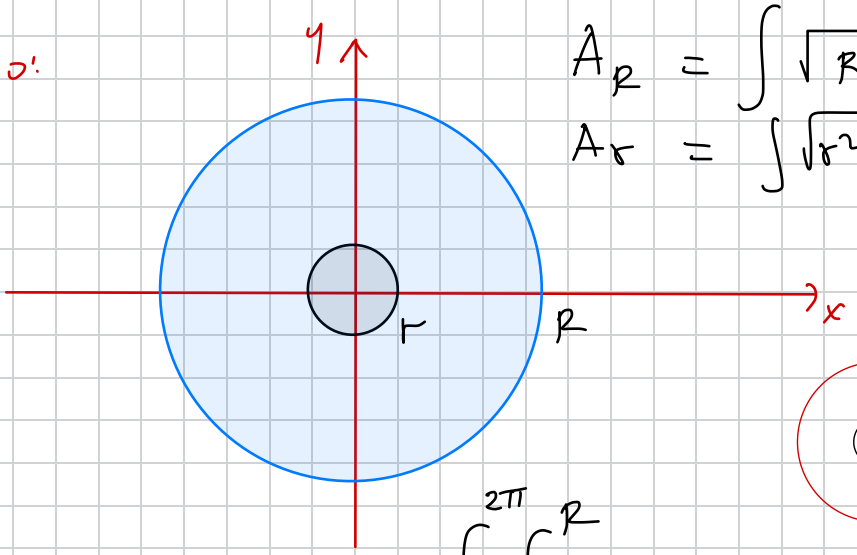
$$\iint_R f(x,y)\,dx\,dy \quad \begin{matrix} dx\,dy \\ dy\,dx \end{matrix} = \iint_{R_{(r,\theta)}} f(r,\theta)\,|J|\,dr\,d\theta$$

$x = r\cos\theta$; $y = r\operatorname{sen}\theta$

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\operatorname{sen}\theta \\ \operatorname{sen}\theta & r\cos\theta \end{vmatrix}$$

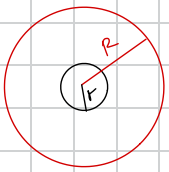
$$|J| = r\cos^2\theta + r\operatorname{sen}^2\theta = r$$

Ejemplo:



$$A_R = \int \sqrt{R^2 - x^2}\,dx$$

$$A_r = \int \sqrt{r^2 - x^2}\,dx$$



$A_R = \pi R^2$

$A_r = \pi r^2$

$A_{total} = A_R - A_r$

$A_{total} = \pi(R^2 - r^2)$

$$\iint_R f(x,y)\,dA \quad \begin{matrix} (dy\,dx) \\ (dx\,dy) \end{matrix} = \int_0^{2\pi}\int_{r_1}^{R} r\,dr\,d\theta$$

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$$= 4\int_0^{\pi/2}\int_{r_1}^{R} r\,dr\,d\theta$$

$$= 4\int_0^{\pi/2}\left(\frac{r^2}{2}\Big|_{r_1}^{R}\right)d\theta$$

$$= 2(R^2 - r_1^2)\cdot\theta\Big|_0^{\pi/2}$$

$$= 2(R^2 - r_1^2)\cdot\pi/2 = \pi(R^2 - r_1^2)$$